UNIT – 2

Intel 8085 Interrupts and DMA: 8085 Interrupts – Software and Hardware Interrupts – 8259 Programmable Interrupt Controller - Data Transfer Techniques – Synchronous, Asynchronous and Direct Memory Access (DMA) and 8237 DMA Controller- 8253 Programmable Interval Timer.

1. Define Hardware and Software Interrupt ***

Hardware: An external device initiates the hardware interrupts and placing an appropriate signal at the interrupt pin of the processor.If the interrupt is accepted then the processor executes an interrupt service routine. Example: TRAP, RST 7.5,6.5& 5.5

Software: The software interrupts are program instructions. These instructions are inserted at desired locations in a program.The 8085 has eight software interrupts from RST 0 to RST 7. The vector address is calculated by Interrupt number * 8 = vector address. Example: For RST 5, 5 * 8 = 40 = 28H

2. Give the vector interrupt in 8085 and their location ***

Interrupt	Vector address
RST 7.5	$003C_H$
RST 6.5	0034_H
RST 5.5	$002C_H$
TRAP	0024_H

3. What are the disadvantages of using 8259? ***

- All request are vectored to memory location on00_H, which is reserved for ROM or EPROM and access to this location is difficult often the system has been designed
- Process of determining the priorities are limited and the extra hardware is required to insert the restart instruction.

4. Uses of TRAP ***

- This interrupt is a non-maskable interrupt. It is unaffected by any mask or interrupt enable
- In sudden power failure, it executes a ISR and send the data from main memory to backup memory.

5. Difference between maskable and non-maskable interrupt***

Maskable Interrupt	Non-Maskable Interrupt
1. This interrupt can be turned off by the programmer	1. It cannot be turned off by the programmer
2. RST 7.5, 6.5, 5.5 are maskable interrupt	2. TRAP is the only non-maskable interrupt.

6. Define Interrupt service routine. ***

Interrupt is an signal send by an external device to the processor so as to repeat processor to perform a particular task or work.

- ISR is used to store information about the interrupts currently being serviced
- OCWs →Operation Control Word

7. Give the priorities of 8085 interrupts

- TRAP bas the highest priority and vectored interrupt
- The RST 7.5 interrupt is a maskable interrupt.
- The RST 6.5 has the third priority whereas RST 5.5 has the fourth priority.
- INTR is the low priority interrupt

8. Difference between synchronous and asynchronous data transfer ***

Synchronous Data Transfer	Asynchronous Data Transfer
1. Simplest data transfer method and can be used when the speed of the microprocessor and the I/O devices matches	1. Data is transferred from the peripheral to the processor or from the processor to the peripheral only after getting a strobe signal from the peripheral.
2. Data can be transferred from processor to peripheral using the OUT instruction	2. Till the handshake signal is received, the processor is in a loop and cannot perform any other task. (i.e.) valuable processor time is wasted.

9. What are the primary features of 8259?

- 8259 manages 8 interrupt requests ($IR_0 – IR_7$)
- 8259 can solve eight levels of interrupt priorities in a variety of modes.
- With additional 8259 devices, the priority scheme can be expanded to 64 levels.
- 8259 is designed to operate only with 8 bit processors. 8259 A is designed to operate only with 8 bits as well as 16 bit processors.

10. What is an interrupt I/O?

The interrupt I/O is a process of data transfer whereby an external device or a peripheral can inform the processor that it is ready for communication and it requests attention

11. Write an instruction to enable all the interrupts in an 8085 system? ***

EI
MVI A,08H .
SIM

12. How the 8327 DMA controller transfers 64K bytes of data per channel with Address lines?

The most significant bits D15 and D14 of the count register are used to specify DMA function and the remaining fourteen bits are used to specify the number of bytes to be transferred.

13. What are the signals used by the DMA controller? ***

The Signals are:

- HLDA
- DMA request
- DMA acknowledge
- AEN – address enable
- ADSTB- address strobe

14. Give the additional features of 8259A controller?

- Input triggering
- Interrupt Status
- Poll Method

15. How the signals of the 8237 are classified? ***

The signals are classified in to two groups.
i. One group of signals are used for interfacing with the MPU
ii. Second group for communicating with the peripherals.

16. How long the INTR pulse stays high?

The INTR pulse can remain high until the interrupt flip-flop is set by the EI instruction in the service routine.

17. Give the three formats of END of Interrupt? ***

- NON-specific EOI command
- Specific EOI command
- Automatic interrupt

18. What are the two modes of DMA execution? ***

Slave Mode, Master mode

19. What do you mean by control logic?

This has two pins. INT as an output, and INTA as an input. The INT isconnected to the interrupt pin of the MPU.

20. Give the commonly used priority modes?

- Fully Nested mode
- Automatic rotation mode
- Specific rotation mode

21. What is RIM? ***

RIM: Read Interrupt Mask Used for three functions
a. To read interrupt mask
b. To identify the pending interrupt
c. To receive serial data

22. What is SIM? ***

SIM: Set interrupt Mask. It is a 1-byte instruction. Used for three functions
a. To set the Mask
b. To reset the flip flop
c. Implement the I/O

23. Give the operating modes of 8259a?

- Fully Nested Mode
- End of Interrupt (EOI)
- Automatic Rotation
- Automatic EOI Mode
- Specific Rotation
- Special Mask Mode
- Edge and level Triggered Mode
- Reading 8259 Status
- Poll command
- Special Fully Nested Mode
- Buffered mode
- Cascade mode

24. List the command words of 8259A?

- Initialization command word &
- Operation command word

25. Name the 6 modes of operations of an 8253 programmable interval timer.***

- Mode 0:interrupt on terminal count
- Mode 1:hardware re -triggerable one-shot
- Mode 2 :rate generator
- Mode3:square wave rate generator
- Mode 4:software triggered strobe
- Mode 5:hardware triggered strobe

26. What is meant by interrupt? **

Interrupt is an external signal that causes a microprocessor to jump to a specific subroutine (ie. Program counter is modified to address specified in the ISR).

Unit II

Intel 8085 Interrupts and DMA: 8085 Interrupts – Software and Hardware Interrupts – 8259 Programmable Interrupt Controller - Data Transfer Techniques – Synchronous, Asynchronous and Direct Memory Access (DMA) and 8237 DMA Controller- 8253 Programmable Interval Timer.

INTERRUPTS

Definition:

Interrupt is a signal send by an external device to the processor so as to request the processor to perform a particular task or work.

Interrupt is a process where an external device can get the attention of the microprocessor.

Analogy of interrupt process:

Assume that you are reading an interesting novel at your desk, where there is a telephone. For you to receive and respond to a telephone call, the following step should occur:

1. The telephone system should be enabled, meaning that the receiver should be on the hook.

2. You should glance at the light at certain intervals to check whether someone is calling.

3. If you see a blinking light, you should pick up the receiver, say hello, and wait for a response. Once you pick up the phone, the line is busy, and no more calls can be received until you replace the receiver.

4. Assuming that the caller is your roommate, the request may be: It is going to rain today. Will you please shut all the windows in my room?

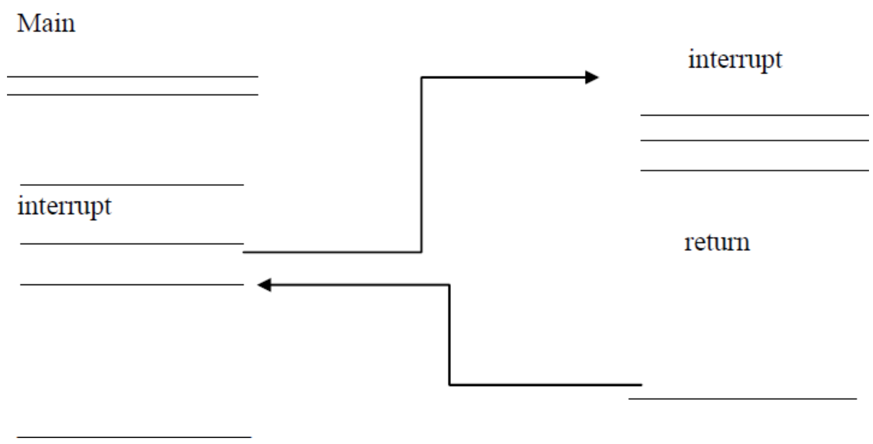
5. You insert a bookmark on the page you are reading.

6. You replace the receiver on the hook.

7. You shut your roommate’s windows.

8. You go back to your book, find your mark, and start reading again.

[Note: steps 6 and 7 may be interchanged, depending on the urgency of the request.]



Classification Of Interrupts:

Interrupts can also be classified into two types:

Maskable Interrupts (Can be delayed or rejected)

Non-Maskable Interrupts (Cannot be delayed or rejected)

Interrupts can also be classified into two types:

Vectored (the address of the service routine is hard-wired)

Non-vectored (the address of the service routine needs to be supplied externally by the device)

An interrupt is considered to be an emergency signal that may be serviced.

- The Microprocessor may respond to it as soon as possible.

What happens when MP is interrupted?

When the Microprocessor receives an interrupt signal, it suspends the currently executing program and jumps to an Interrupt Service Routine (ISR) to respond to the incoming interrupt.

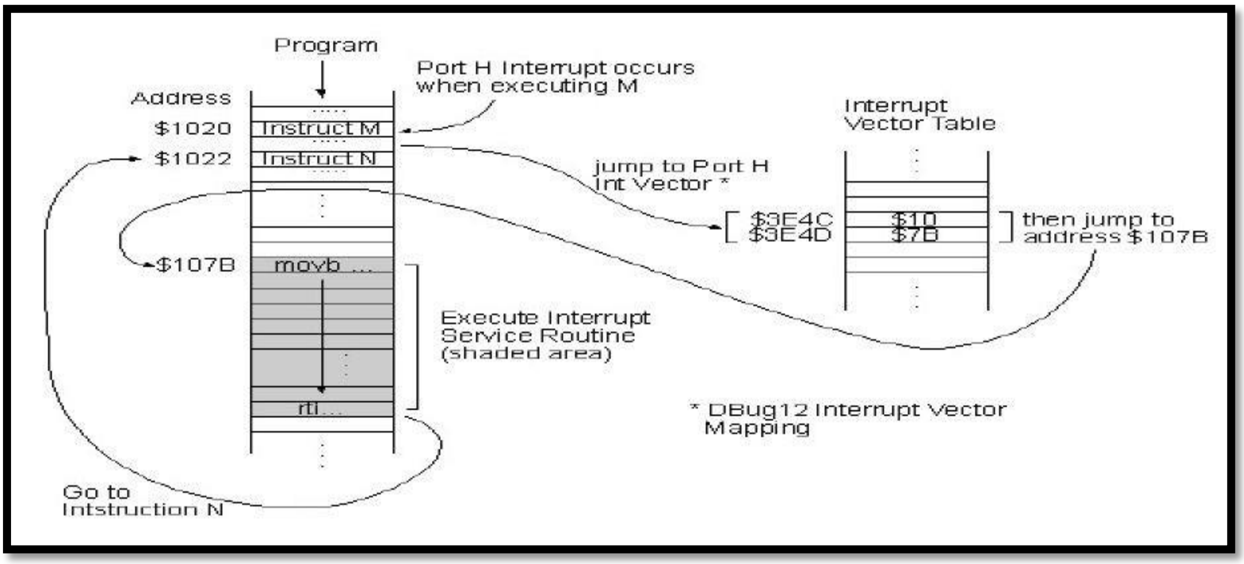
Each interrupt will most probably have its own ISR.

RESPONDING TO INTERRUPTS:

Responding to an interrupt may be immediate or delayed depending on whether the interrupt is maskable or non-maskable and whether interrupts are being masked or not.

There are two ways of redirecting the execution to the ISR depending on whether the interrupt is vectored or non-vectored.

- *Vectored:* The address of the subroutine is already known to the Microprocessor.
- *Non Vectored:* The device will have to supply the address of the subroutine to the Microprocessor.



1. THE 8085 INTERRUPTS:

When a device interrupts, it actually wants the MP to give a service which is equivalent to asking the MP to call a subroutine. This subroutine is called ISR (Interrupt Service Routine)

- The 'EI' instruction is a one byte instruction and is used to Enable the non-maskable interrupts.
- The 'DI' instruction is a one byte instruction and is used to Disable the non-maskable interrupts.
- The 8085 has a single Non-Maskable interrupt.
- The non-maskable interrupt is not affected by the value of the Interrupt Enable flip flop.

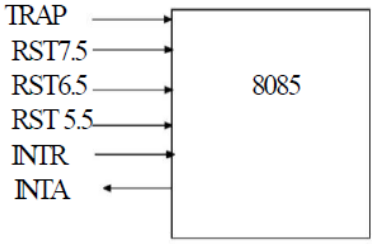
The 8085 Has 5 Interrupt Inputs:

- The INTR input
    - ✓ The INTR input is the only non-vectored interrupt
    - ✓ INTR is maskable using the EI/DI instruction pair

- RST 5.5, RST 6.5, RST 7.5 are all automatically vectored
    - ✓ RST 5.5, RST 6.5, and RST 7.5 are all maskable

- TRAP is the only non-maskable interrupt in the 8085
    - ✓ TRAP is also automatically vectored

Interrupt name	Maskable	Vectored
INTR	Yes	No
RST 5.5	Yes	Yes
RST 6.5	Yes	Yes
RST 7.5	Yes	Yes
TRAP	No	Yes



8085 Interrupts

Interrupt Vectors And The Vector Table:

- An interrupt vector is a pointer to where the ISR is stored in memory.
- All interrupts (vectored or otherwise) are mapped onto a memory area called the Interrupt Vector Table (IVT).

  ✓ The IVT is usually located in memory page 00 (0000H - 00FFH).

  ✓ The purpose of the IVT is to hold the vectors that redirect the microprocessor to the right place when an Interrupt arrives

Example: Let, a device interrupts the Microprocessor using the RST 7.5 interrupt line.

Because the RST 7.5 interrupt is vectored, Microprocessor knows, in which memory location it has to go using a call instruction to get the ISR address. RST7.5 is knows as Call 003Ch to Microprocessor. Microprocessor goes to 003C location and will get a JMP instruction to the actual ISR address. The Microprocessor will then, jump to the ISR location

The 8085 Non-Vectored Interrupt Process:

1. The interrupt process should be enabled using the EI instruction.

2. The 8085 checks for an interrupt during the execution of every instruction.

3. If INTR is high, MP completes current instruction, disables the interrupt and sends INTA (Interrupt acknowledge) signal to the device that interrupted.

4. INTA allows the I/O device to send a RST instruction through data bus.

5. Upon receiving the INTA signal, MP saves the memory location of the next instruction on the stack and the program is transferred to 'call' location (ISR Call) specified by the RST instruction.

6. Microprocessor Performs the ISR.

7. ISR must include the 'EI' instruction to enable the further interrupt within the program.

8. RET instruction at the end of the ISR allows the MP to retrieve the return address from the stack and the program is transferred back to where the program was interrupted.

The 8085 Recognizes 8 Restart Instructions:

RST0 - RST7.

Each of these would send the execution to a predetermined hard-wired memory location:

Restart Instruction	Equivalent to
RST0	CALL 0000H
RST1	CALL 0008H
RST2	CALL 0010H
RST3	CALL 0018H
RST4	CALL 0020H
RST5	CALL 0028H
RST6	CALL 0030H
RST7	CALL 0038H

1.2 SOFTWARE INTERRUPTS OF 8085:

1. The software instructions are program instructions. When a software interrupt instruction is executed, the processor executes an interrupt service subroutine (ISR) stored in the vector address of that software interrupt instruction.

2. The software interrupts of 8085 are RSR 0, RST 1, RST 2, RST 3, RST 4, RST 5, RST 6 and RST 7. The software interrupts of 8085 are vectored interrupts. The software interrupts cannot be masked and they cannot be disabled.

3. The vector addresses of software interrupt are given in the table.

4. The software interrupt instructions are included at the appropriate (or required) place in the main program.

5. When the processor encounters the software instruction, it pushes the content of the PC (program counter) to stack.

6. Then loads the vector address in PC and starts executing an ISR stored in this address.

7. The last instruction of ISR will be RET instruction. When the RET instruction is executed, the

8. Processor POP the content of the top of stack to PC.

Hence the processor control returns to the main program after receiving the interrupt.

1.3 HARDWARE INTERRUPTS OF 8085:

INTERRUPTS	VECTOR ADDRESS
RST 7.5	003CH
RST 6.5	0034H
RST 5.5	002CH
TRAP	0024H

The hardware interrupts of 8085 are initiated by an external device by placing an appropriate signal at the interrupt pin of the processor.

- The processor keeps on checking the interrupt pins at the second T-state of last machine cycle of every instruction.
- If the processor finds a valid interrupt signal and if the interrupt is unmasked and enabled; then the processor accepts the interrupt. The acceptance of the hardware interrupt is acknowledged by sending an INTA signal to the interrupting device.
- When the interrupt is accepted; the processor saves the content of PC into the stack.
- The hardware interrupts of 8085 are TRAP, RST 7.5, RST 6.5, RST 5.5 and INTR. except INTR all are vectored interrupts.
- In vectored interrupts the address to which the program control is transferred is fixed by the manufacturer.
- The vector addresses of hardware interrupts are given in table.
- TRAP is edge and level sensitive. Hence to initiate TRAP, the interrupt signal has to make a low to high transition and then it has to remain high until the interrupt is recognized.
- The RST 7.5 is edge sensitive. , and in order to initiate it ,the interrupt signal has to make a low to high transition and then it need not remain high until the interrupt is recognized.

- The RST 7.5, RST 6.5, RST 5.5 and INTR are level sensitive. Hence these interrupts should remain high, until it is recognized.
- The Trap is non maskable interrupt and RST 7.5, RST 6.5, RST 5.5 are maskable interrupt using SIM instruction. The status of maskable interrupts can be read into the accumulator by executing RIM instruction.
- All the interrupts except TRAP are disabled when the processor is resettled or can also be disabled by executing DI instruction.
- In order to enable them the processor has to execute EI instruction.

1.4 MULTIPLE INTERRUPTS & PRIORITIES:

- The microprocessor can only respond to one signal on INTR at a time. Therefore, we must allow the signal from only one of the devices to reach the microprocessor.
- We must assign some priority to the different devices and allow their signals to reach the microprocessor according to the priority.

The Priority Encoder

- The solution is to use a circuit called the priority encoder (74LS148).

  – This circuit has 8 inputs and 3 outputs.

  – The inputs are assigned increasing priorities according to the increasing index of the input.

- Input 7 has highest priority and input 0 has the lowest.

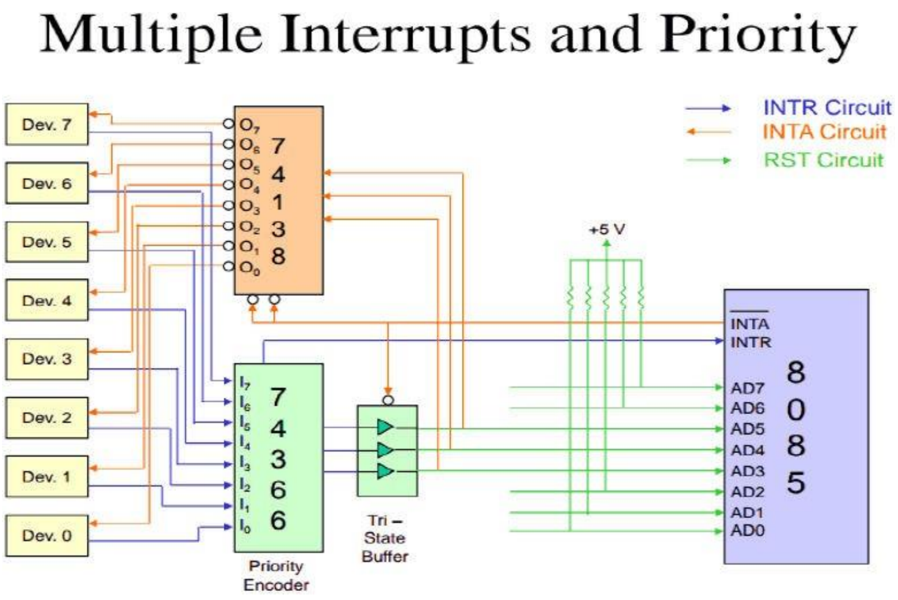
  – The 3 outputs carry the index of the highest priority active input.

  – Below figure shows how this circuit can be used with a Tri-state buffer to implement an interrupt priority scheme.

Multiple Interrupts & Priorities:

- Note that the opcodes for the different RST instructions follow a set pattern.
  - Bit D5, D4 and D3 of the opcodes change in a binary sequence from RST 7 down to RST 0.
  - The other bits are always 1.
  - This allows the code generated by the 74366 to be used directly to choose the appropriate RST instruction.

The one drawback to this scheme is that the only way to change the priority of the devices connected to the 74366 is to reconnect the hardware.



The 8085 Maskable/Vectored Interrupts:

- The 8085 has 4 Masked/Vectored interrupt inputs.

  – RST 5.5, RST 6.5, RST 7.5

  - They are all maskable.
  - They are automatically vectored according to the following table:

Interrupt	Vector
RST 5.5	002CH
RST 6.5	0034H
RST 7.5	003CH

The vectors for these interrupt fall in between the vectors for the RST instructions. That's why they have names like RST 5.5 (RST 5 and a half).

Masking RST 5.5, RST 6.5 And RST 7.5:

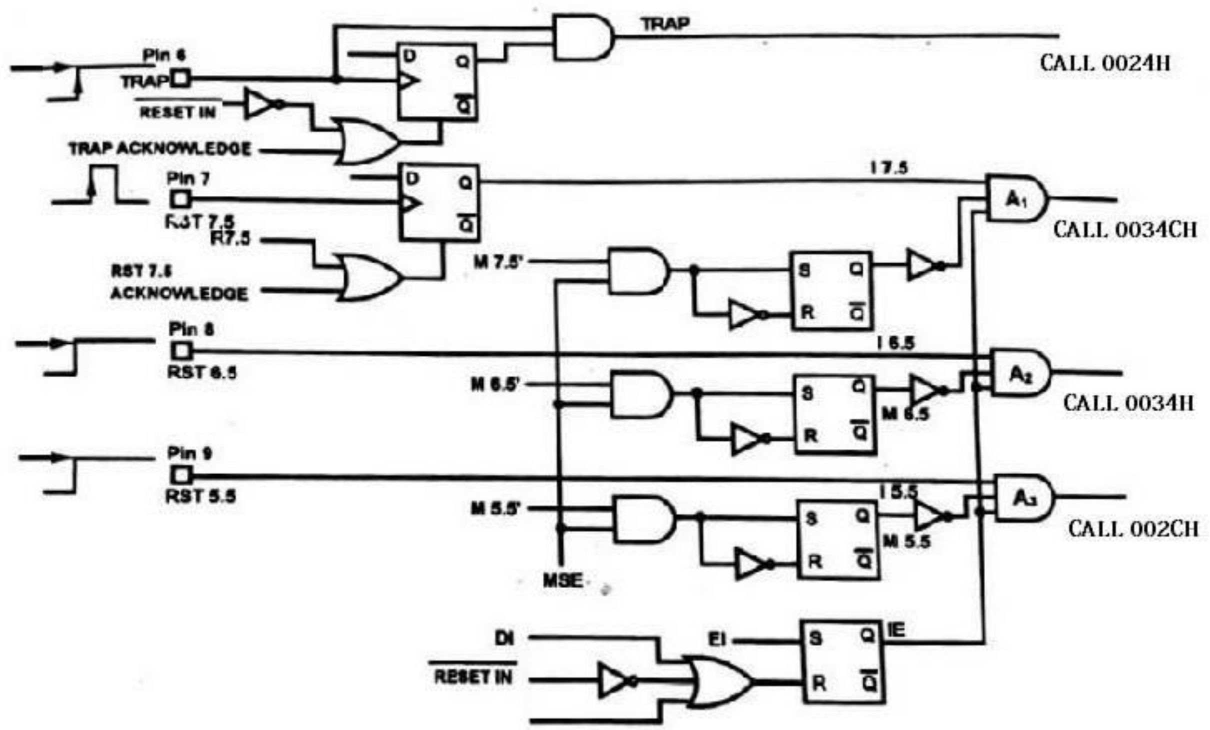
- These three interrupts are masked at two levels:

  1. Through the Interrupt Enable flip flop and the EI/DI instructions.

     - The Interrupt Enable flip flop controls the whole maskable interrupt process.

  2. Through individual mask flip flops that control the availability of the individual interrupts.

- These flip flops control the interrupts individually.



Maskable Interrupts and vector locations

The 8085 Maskable/Vectored Interrupt Process:

1. The interrupt process should be enabled using the EI instruction.

2. The 8085 checks for an interrupt during the execution of every instruction.

3. If there is an interrupt, and if the interrupt is enabled using the interrupt mask, the microprocessor will complete the executing instruction, and reset the interrupt flip flop.

4. The microprocessor then executes a call instruction that sends the execution to the appropriate location in the interrupt vector table.

5. When the microprocessor executes the call instruction, it saves the address of the next instruction on the stack.

6. The microprocessor jumps to the specific service routine.

7. The service routine must include the instruction EI to re-enable the interrupt process.

8. At the end of the service routine, the RET instruction returns the execution to where the program was interrupted.

2. 8259 PROGRAMMABLE INTERRUPT CONTROLLER

The following are the features of 8259A:

1. 8259A handles up to 8-vectored priority interrupts for the CPU
2. It is cascadable for up to 64-vectored priority interrupts without additional circuiting.
3. The priority modes can be changed or reconfigured dynamically at any time during the main program.
4. 8259 can be used with 8080/8085 or 8086/8088 microprocessors.
5. The various interrupt modes it can operate are:
    - Fully nested mode
    - Rotating priority mode
    - Special mask mode
    - Polled mode
6. 8259A supports both edge & level triggered mode of interrupted.
7. The data bus can be buffered.
8. The AEOI can be programmed.
9. The CALL address interval can be programmed to either 4 or 8.

Pin configuration:

- 8259A is packed in a 28 pin DIP, using NMOS technology and requires a single +5 V supply.
- Circulating is static, requiring no check input.
- The pin configuration of 8259A is illustrated in fig.

D0-D7 (Data Bus)

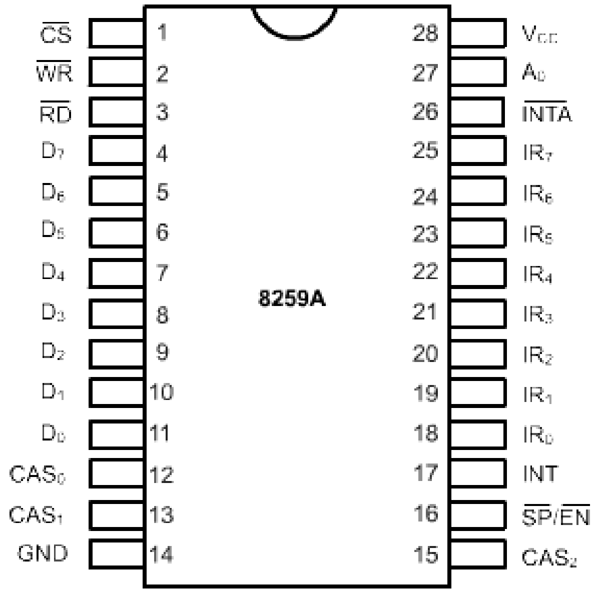
- Bidirectional data is used to transfer control, status and interrupt vector information.

A0 (Address Line)

- This pin acts in conjunction with CS,WR and RD pins.
- It is used by 8259A to decipher various command words.
- The CPU writes to initialize command words and reads the status.
- It is typically connected to the CPU. A1 reads the address line of 8086.

CS (Chip Select)

- A low on this pin enables RD and WR communication between the CPU and the 8259A.INTA functions are independent of CS.



Pins	Functions
D_7 - D_0	Data Bus (Bidirectional)
A_0	Command Select Address
$\overline{RD}$	Read Input
$\overline{WR}$	Write Input
$\overline{CS}$	Chip Select
CAS_1 - CAS_3	Cascade Lines
$\overline{SP}/\overline{EN}$	Slave Program/Enable Buffer
INT	Interrupt Output
$\overline{INTA}$	Interrupt Acknowledgement Input
IR_0 - IR_7	Interrupt Request Inputs

BLOCK DIAGRAM

The internal block diagram of 8259 contains the 4 sections, they are:

1. Interrupts and control logic sectionData bus buffer
2. Read/Write control logic section
3. Cascade Buffer/Comparator Section

Interrupts and Control Logic Section:

This section consists of the following registers:

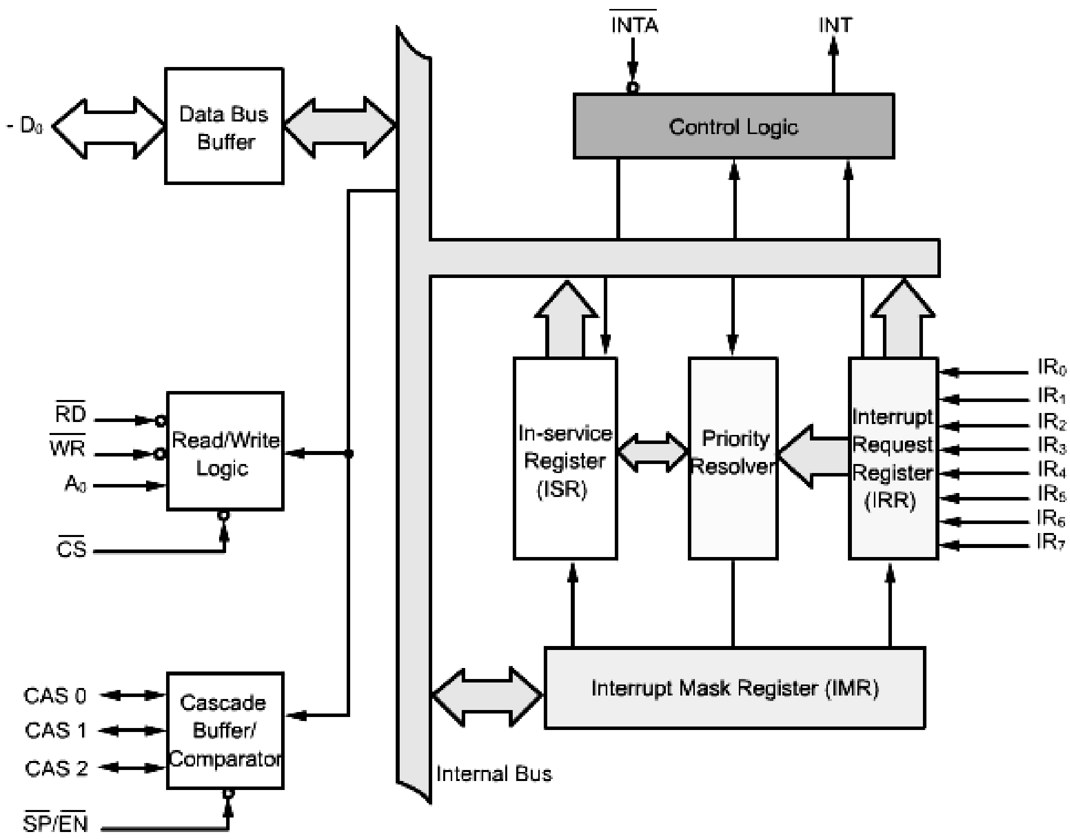
(a) Interrupt request Register (IRR)

(b) In-Service Register (ISR)

(c) Priority Resolver

(d) Interrupt Mask Register (IMR)

(e) Control logic Block



Block Diagram

Data Bus Buffer

- This 3-state, bidirectional 8-bit data buffer is used to interfere the 8259A to the system data bus.
- Control words status information are transferred through the data bus buffer.
- INTA functions are independent of CS.

Read/Write Logic

- The function of this block is to accept commands from the CPU.
- Command Word Registers (OCW registers) which are programmed by the processor to set up the 8259, and to operate This section contains the initialization Command Word Registers (ICW registers) and the Operation it in various modes.
- This section also accepts Read commands from the CPU to permit the CPU to read status words.
- When the address line Ao is at logic 0, the controller is selected to write a command or read a status.
- The chip select logic and Ao determine the port address of the controller.
- The operation command word (OCW) register which store the various control formats which define the device operation.
- This function block also allows the status of the 8359Ato be transferred onto the data bus.

The pins associated with this section are described below:

CHIP SELECT:

- This is an active low input which is used to select the device

WRITE:

- This is an active low input and is used to write OCWs and ICWs onto the 8259

READ:

- This is also an active low input.
- It is used by the CPU to read the status of the IRR, ISR, IMR, or the interrupt level.

Cascade Buffer / Comparator

- This function block stores and compares the ID (identification code) of all the 8259A's used in the system.
- The 8259 can be cascaded with other 8259s in order to expand the interrupt handling capacity to sixty-four levels.
- In such a case, the former is called a master and the latter are called slaves.
- This block is used to expand the number of interrupt levels.
- The associated 3 I/O pins (CAS0-CAS2) are outputs when the 8259A is used as a master and are inputs when the 8259A is used as a slave.
- As a master, the 8259A sends the ID of the interrupting slave device on to the CAS0-CAS2 lines.

The slave thus selected will output the pre-programmed subroutine address onto the data bus in response to the INTA pulses from the CPU.

Control Logic

- The logic blocks controls the overall operation of the controller.
- This block has an input and output line.
- The 8259, after resolving its input interrupt request priorities, places an interrupt request to the processor on the INT output.
- This is directly connected to the cpu interrupt input.
- It generates interrupt to the microprocessor and receive INTA signal.
- It enables the data bus buffer to send the required information when $\overline{\text{INTA}}$ signal is obtained.

Interrupt request register (IRR)

- IRR is used to store all the interrupt levels which are requesting service.
- IRR has eight input lines for interrupts.
- The eight interrupt inputs set corresponding bits of the Interrupt request register.

- When these lines go to HIGH, the requests are stored in the register

Priority Resolver (PR)

- The logic block determines the priorities of the bits set in the IRR.
- The highest priority is selected and strobed into the corresponding bit of the ISR in response to the pulse.

Interrupt Mask Register (IMR)

- The IMR stores the interrupt mask information (i.e., whether each interrupt has to be enabled/disabled).
- This register can be programmed by an OCW to store the bits which mask interrupts.
- The 8 bits of the IMR are used to disable or enable individual interrupt inputs; writing 0 in the corresponding bit location enables the interrupt.
- An interrupt, which is masked by software (by programming the IMR), will not be recognized and serviced even if it sets the corresponding bit in the IRR.
- The IMR operates on the IRR. Masking of a higher priority input will not affect the interrupt requests lines of lower priority interrupts.

Interrupt Service Register (ISR)

- The ISR is used to store all the interrupt levels which are being serviced.

Interrupt Operations:

- First, the 8259 should be initialized by placing control words in the control register.
- It requires two types of control words:

  (1)Initialization command words

  (2) Operational command words

Priority Interrupt Modes:

- Many types of priority modes are available under software control in the 8259.
- They can be changed dynamically during the program by writing appropriate command words.

Commonly used priority modes are given below:

1. FULLY NESTED MODE:

- This is a general purpose mode.
- In this mode all IRs (Interrupt Requests) arranged from highest to lowest IRo has the highest priority and IR7 has the lowest priority.

2. AUTOMATIC ROTATION MODE:

- In this mode, a device after being serviced, receives the lowest priority .

3. SPECIFIC ROTATION MODE:

- This mode is similar to the automatic rotation maode, expect that the user can select any IR for the lowest priority , thus fixing all other priorities.

END OF INTERRUPT

- After the completion of the interrupt service, the corresponding ISR bit needs to be reset to update the information in the ISR.
- This is called the end-of-interrupt (EOI) command.
- It can be issued three formats.

1. Non Specific EOI command

When this command is sent to the 8259 it resets the highest priority ISR bit.

2. Specific EOI command

This command specifies, which ISR bit to reset. 3. AUTOMATIC EOI

- In this mode no command is necessary.
- The major drawback with this mode is that the ISR does not have information on which IR is being serviced .
- Thus nay IR can interrupt the service routine, irrespective of its priority , if the interrupt enable Flip-Flop is set.

VECTORING DATA FORMATS

- The eight interrupts levels generate CALLs to eight equally spaced locations in memory.
- These locations can be programmed to be spaced at an intyerval of four or eight locations.
- The vectoring table will therefore be either a page of thirty-two bytes , or a page of sixty- four bytes.

INITIALIZATION COMMAND WORD (ICW1)

- ICW 1 is used to give the information about single or multiple 8259s in the system.
- 4 or 8 bytes of interval between the interrupt vector locations .
- The address bit A7- A5 of the CALL instruction.

IC4

- This is used toi specify whether ICW4 is required or not required.
- If it is set to 0, the 8259 is set in the non-buffered mode.

SIGNAL

- This is used to inform the the 8259 if it is the only 8259 in the system, or if additional 8259s are present.

ADI

- This sets the CALL address interval to either four or eight

LTIM

- This bit determines if the interrupts are to be recognized in the level-triggered mode or in the edge - triggered mode.

A5 –A7

- A0 – A4 of the vectoring address are automatically inserted the 8259 for all the IR inputs for an interval spacing of four.
- A0 – A5 are inserted automatically for an interval spacing of eight.

INITIALIZATION COMMAND WORD2 (ICW 2)

- A write command following ICW1 with A0=1, is interpreted as ICW2.
- The format of the byte to be loaded as ICW2
- This is used to lad the high-order byte of the interrupt vector address of all the interrupts.
- This byte is common for all interrupts.

OPERATION COMMAND WORDS (OCWS)

- After initialization, the 8259 is ready to process interrupt requests.
- Operation Command Words are used to change manner of processing the interrupts during operation.
- They may be located anytime after the 8259s initialization to dynamically after the priority modes.

4. DATA TRANSFER TECHNIQUES

The data transfer technique refers to the method of data transfer between the processor and peripheral devices. In a typical microcomputer, data transfer takes place between any two devices:

- Microprocessor and memory

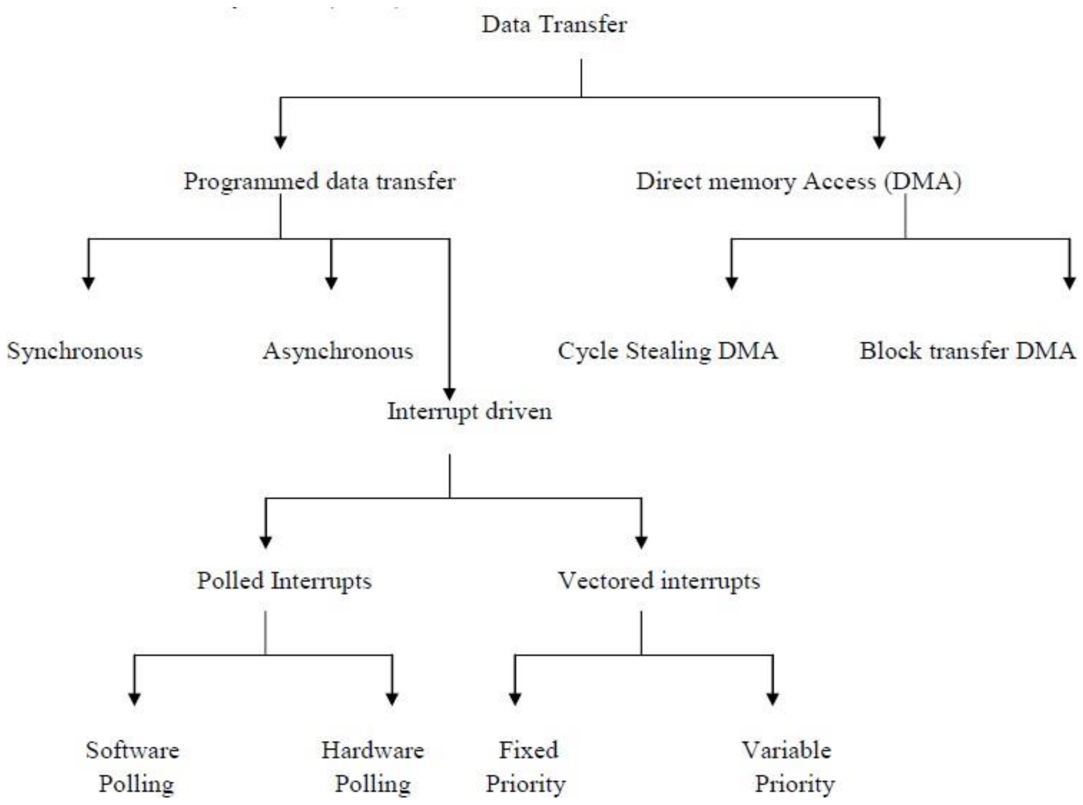
- Microprocessor and I/O devices
- Memory and I/O devices

For effective data transfer between these devices, the timing parameters of the devices should be matched.

TYPES OF DATA TRANSFER:-

The data transfer techniques have been broadly classified into the following two categories;

- Programmed data transfer
- Direct memory access (DMA) data transfer



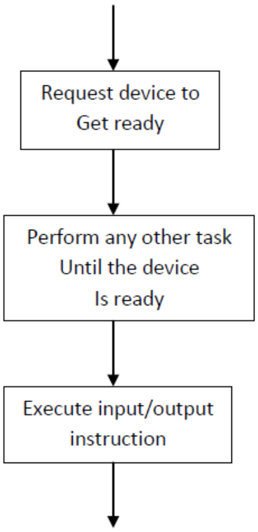
PROGRAMMED DATA TRANSFER:

In programmed data transfer, a memory resident request the device for data transfer to or from one of the processor register. It is used when relatively small amount of data are to be transferred. They can be further classified into the following three types;

- Synchronous data transfer
- Asynchronous data transfer
- Interrupt driven data transfer

SYNCHRONOUS DATA TRANSFER:

The synchronous data transfer scheme is the simplest of all data transfer schemes. In this scheme the processor does not check the readiness of the devices. The I/O device or peripheral should have matched timing parameters. Whenever data is to be obtained from the device or transferred to the device, the user program can issue a suitable instruction for the device. At the end of he execution of this instruction, the transfer would have been completed.



Synchronous data Transfer scheme

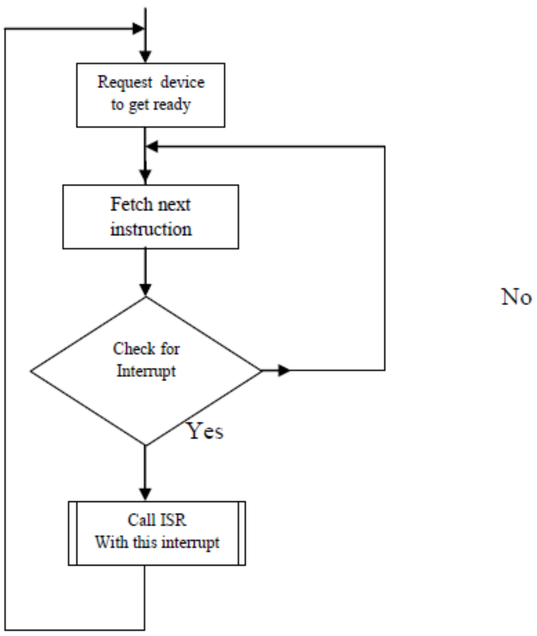
ASYNCHRONOUS DATA TRANSFER:

The asynchronous data transfer scheme is employed when the speed of the processor and I/O devices does not match. In this scheme the processor sends a request to the device for read/write operation. Then the processor keeps on polling the status of the devices. Once the device is ready, the processor executes a data transfer instruction to complete the process.

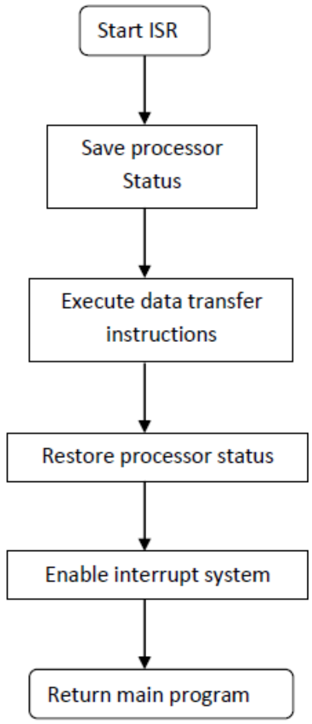
INTERRUPT DRIVEN DATA TRANSFER:

The interrupt driven data transfer scheme is the best method of data transfer for efficient utilization of processor time. In this scheme, the processor first initiates the I/O device for data transfer. After initiating the device, the processor will continue the execution of instructions in the program. Also at the end of every instruction the processor will check for a valid interrupt signal. If there is no interrupt then the processor will continue the execution.

When the I/O device is ready, it will interrupt the processor. On receiving an interrupt signal the processor will complete the current instruction execution and save the processor status in stack. Then the processor calls an Interrupt Service Routine (ISR) to service the interrupting device. At the end of ISR, the processor status is retrieved from the stack and the processor starts executing its main program.



Main program execution sequence.



ISR execution sequence

4. DIRECT MEMORY ACCESS (DMA):

Normally the data transfer from memory to I/O device or I/O device to memory can be achieved only through microprocessor. When data has to be transferred from memory to I/O device, first the processor sends address and control signals to memory to read the data from memory. Then the processor send address and control signals to I/O device to write data to I/O device. Similarly, the data can be transferred from I/O device to memory.

In the data transfer method described above, the data cannot be directly transferred between memory and I/O devices, even though they are connected to common bus. This process is inevitable, because the processor cannot simultaneously select two devices. Hence a scheme called DMA has been developed in which the I/O device can access the memory directly for data transfer.

The DMA data transfer will be useful to transfer large amount of data between memory and I/O device in a short time.

The modes of DMA operations are two types;

- Burst mode
- Cycle stealing

BURST MODE DATA TRANSFER:

As bus control is granted to the device controller, it continues with the controller till the data transfer is completed. After all the data has been transferred, the device interrupts the processor to indicate the completion to the user program. During this period the microprocessor is idling and it is in hold state.

The microprocessor exits from this state only after an interrupt is received or after the DMA request is withdrawn by the peripheral device. The duration of HOLD depends on I/O device speed, the memory speed and the number of bytes transferred. This type of DMA transfer is known as burst mode data transfer.

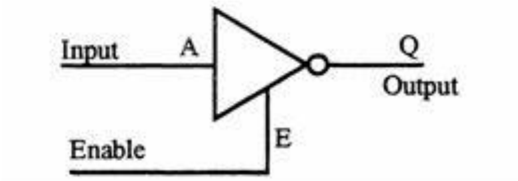
CYCLE STEALING DATA TRANSFER:

The I/O device uses the concept of cycle stealing. The I/O device requests the processor for DMA cycle. When request is granted a byte or a word is transferred and DMA request is withdrawn. After sometime, when the device is again ready for data transfer, it repeats the above process.

Finally when the last data byte has been transferred, the device interrupts the processor indicating the end of the requested I/O operation. This type of DMA access is cycle stealing data transfer.

Direct Memory Access (DMA)

Tri state devices:



- 3 output states are high & low states and additionally a high impedance state.
- When enable E is high the gate is enabled and the output Q can be 1 or 0 (if A is 0, Q is 1, otherwise Q is 0). However, when E is low the gate is disabled and the output Q enters into a high impedance state.

E	A	Q	State
1(high)	0	1	High
1	1	0	Low
0(low)	0	0	High impedance
0	1	0	High impedance

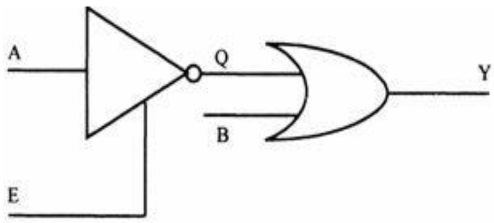


Fig (a) - Pin Diagram of 8085 *Fig (b) - logical schematic of Pin diagram.*

- For both high and low states, the output Q draws a current from the input of the OR gate.
- When E is low, Q enters a high impedance state; high impedance means it is electrically isolated from the OR gate's input, though it is physically connected. Therefore, it does not draw any current from the OR gate's input.

- When 2 or more devices are connected to a common bus, to prevent the devices from interfering with each other, the tristate gates are used to disconnect all devices except the one that is communicating at a given instant.
- The CPU controls the data transfer operation between memory and I/O device. Direct Memory Access operation is used for large volume data transfer between memory and an I/O device directly.
- The CPU is disabled by tri-stating its buses and the transfer is effected directly by external control circuits.
- HOLD signal is generated by the DMA controller circuit. On receipt of this signal, the microprocessor acknowledges the request by sending out HLDA signal and leaves out the control of the buses. After the HLDA signal the DMA controller starts the direct transfer of data.

READY (input)

- Memory and I/O devices will have slower response compared to microprocessors.
- Before completing the present job such a slow peripheral may not be able to handle further data or control signal from CPU.
- The processor sets the READY signal after completing the present job to access the data.
- The microprocessor enters into WAIT state while the READY pin is disabled.

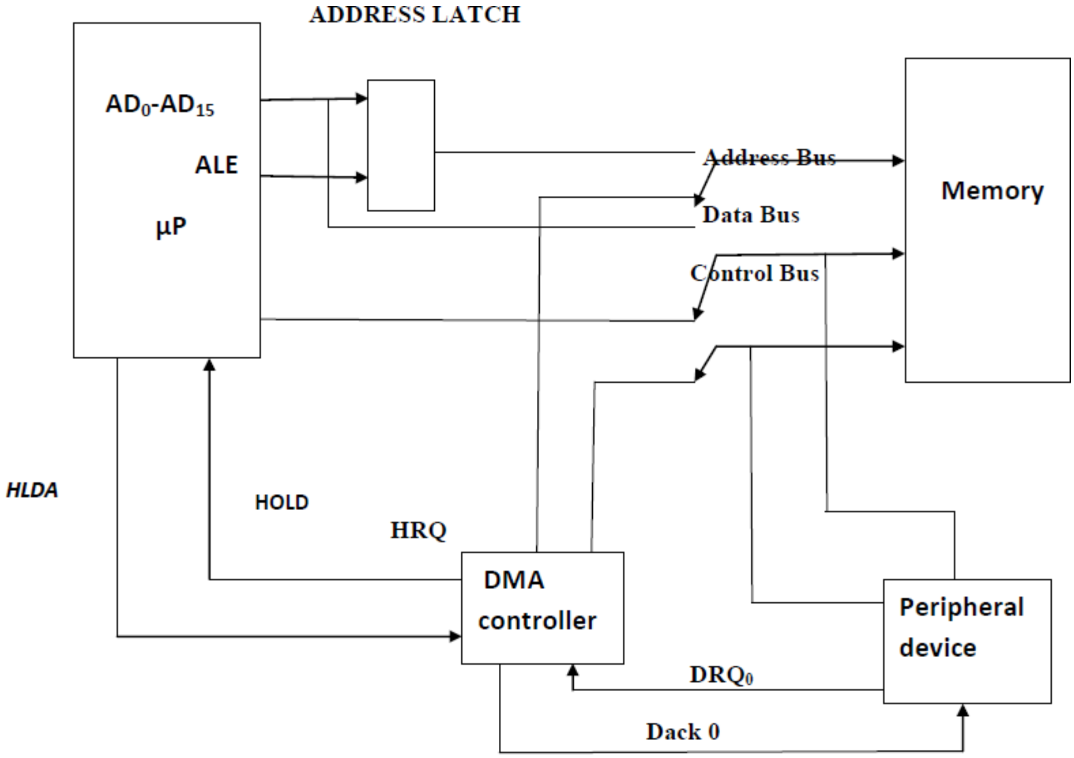
Single Bit Serial I/O ports:

- SID (input) - Serial input data line
- SOD (output) - Serial output data line
- These signals are used for serial communication.

DMA CONTROLLER

To transfer the data at a faster rate, the CPU is isolated and transfer of data is affected between the memory and the peripheral directly. This I/O technique is called Direct Memory Access (DMA).

Direct memory access operation (DMA):



- The DMA operation can be done with three switches.
- Normally, the switch positions are such that the memory and peripheral devices are connected to the CPU, i.e., the data lines, address lines and the control lines of the memory and I/O devices are connected to the CPU.
- Whenever the DMA operation is to be carried out, the CPU I totally isolated and the address lines and control lines are taken over by the DMA controller circuitry.
- To actuate the DMA operation, the following sequence is carried out
  - The device which needs data transfers the device and the memory has to send DMA request (DRQ) to the DMA controller.
  - The DMA controller raises the Hold request (HRQ) line and it is connected to the Hold signal input of the µp
  - The µp tristate all the address lines, data lines and control lines and acknowledges the Hold input signal through Hold Acknowledge (HLDA) output signal.
  - The HLDA signal is connected to the DMA controller. Once it becomes active, then DMA controller takes care of direct data transfer operation.

- It sends acknowledgement signal (DACK)
- To the device which requested for DMA operation to enable the device for data transfer.
- DMA operation is carried out by sending suitable address to the memory and suitable control signals to transfer a byte of data. Then it increments the address and send control signals and so on.
- Before actually carrying out the DMA operation, the DMA controller should know

    A. The starting address of the memory location.

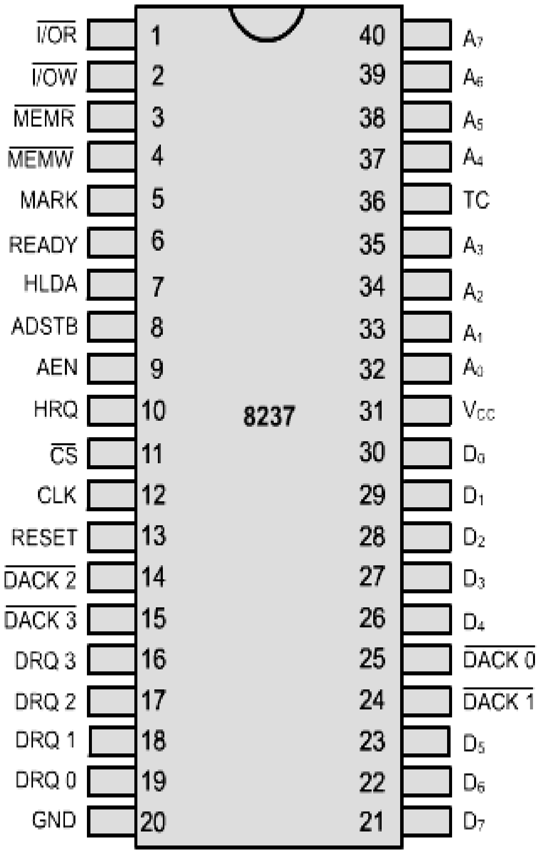
    B. Number of bytes to be transferred.

    C. Whether the transfer is from memory to I/O or from I/O to memory.

6. PROGRAMMABLE 8237 DMA CONTROLLER

Features

- In DMA, data can be transferred between the memory and an external device without utilizing the microprocessor.
- DMA is typically used to transfer large volume of data between memory and an external device.
- A DMA read operation transfers data from memory to an external device.
- A DMA write operation transfers data from an external device to memory.
- The device called DMA controller is used to perform read and write operations in the same manner as the processor.
- Therefore, the DMA controller is actually a special-purpose microprocessor whose only task is to perform high-speed data transfers between memory and an external device.
- The INTEL 8237 DMA controller is a 40 pin programmable device.
- The 8237 has four independent DMA channels.
- One 8237 can provide DMA transfers to four external devices.
- Data transfers begin with DMA request lines DREQ 0 - DREQ 3.
- DREQ 0 has the highest priority and DREQ 3 has the lowest priority.
- DACK 0 – DACK 3 signals are used to acknowledge a DMA channel request.
- After being initialized by the MPU, the 8237 takes control of the bus in order to perform the DMA operation.
- Data bits are then transferred between a peripheral or external device without involving the microprocessor.
- The basic pin configuration of the 8237 is shown in the figure.



Pins	Functions
D_7 - D_0	Data Bus
A_0 - A_7	Address Bus
$\overline{\text{I/OR}}$	I/O Read
$\overline{\text{I/OW}}$	I/O Write
$\overline{\text{MEMR}}$	Memory Read
$\overline{\text{MEMW}}$	Memory Write
CLK	Clock Input
RESET	Reset Input
READY	Ready
HRQ	Hold Request (to 8080A)
HLDA	Hold Acknowledge(from 8080A)
AEN	Address Enable
ADSTB	Address Strobe
TC	Terminal Count
MARK	Modulo 128 Mask
DRQ 3 - DRQ 0	DMA Request Input
$\overline{\text{DACK 3}}$ - $\overline{\text{DACK 0}}$	DMA Acknowledge Out
$\overline{\text{CS}}$	Chip Select
V_{CC}	+5 volts
GND	Ground

Need for 8212 and signal ADSTB

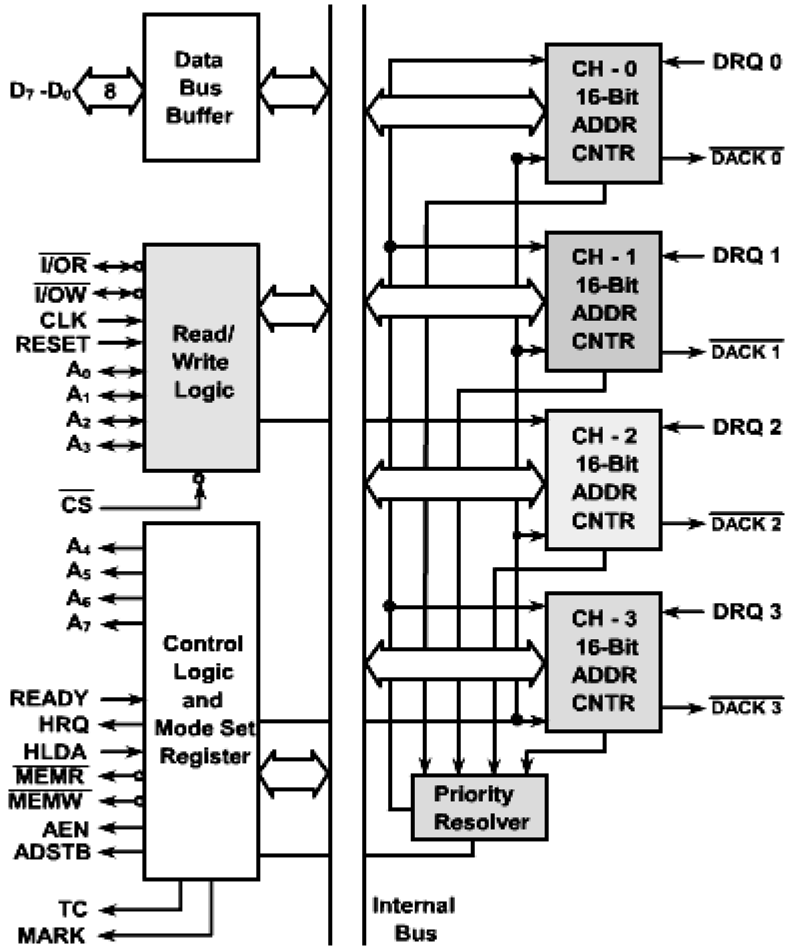
- The 8237 has eight address line, but requires 16 address lines to address a memory location.
- The additional eight lines are generated by using the signal ADSTB to strobe a high-order memory address into 8212 from the data bus.
- A latch, such as the 74LS373 can replace the 5212.

Signal AEN (address enable)

- The AEN output signal is used to disable the system data bus and control bus.
- The signal is necessary to switch the 8237 from the slave mode to the master mode.

Block diagram of INTEL IC 8237

The block diagram of Intel 8237 DMA controller is shown in figure.



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- DACK 0 – DACK 3 signals are used to acknowledge a DMA channel request.
- After being initialized by the MPU, the 8237 takes control of the bus in order to perform the DMA operation.
- Data bits are then transferred between a peripheral or external device without involving the microprocessor.
- Data bus buffer
- The data bus lines are bi-directional three state signals connected to the system data bus.
- The outputs are enabled in the program condition during the I/O read to output the contents of an address register, a status register, the temporary register or a word count register to the CPU.
- The outputs are disabled and the inputs are read during an I/O write cycle when the CPU is programming the 8237A control registers.
- During DMA cycles the most significant 8 bits of the address are output on to the data bus to be stored into an external latch by ADSTB.

- In memory to memory operations, data from memory comes into the 8237A on the data bus during the read from memory transfer.
- In the write to memory transfer, the data bus outputs place the data into the new memory location.

Read / write logic

- The 8237 has eight address lines, but requires 16 address lines to address a memory location.
- The additional eight lines are generated by using the signal ADSTB to strobe a high-order memory address into 8212 from the data bus.
- A latch, such as the 74LS373 can replace the 8212.

I/OR (I/O Read)

- I/O read is a bi-directional active low three state line.
- In the idle cycle, it is an input control signal used by the CPU to read the control registers.
- In the active cycle, it is an output control signal used by the 8237A to access data from a peripheral during DMA write transfer.

I/OW (I/O write)

- I/O write is a bi-directional active low three state line.
- In the idle cycle, it is an input control signal used by the CPU to load information into the 8237A.
- In the active cycle, it is an output control signal used by the 8237A to load data to the peripheral during a DMA read transfer.

CLK (Clock input)

- Clock input controls the internal operations of the 8237A and its rate of data transfers.
- The input may be driven at up to 3 MHZ for the standard 8237A and up to 5 MHZ for the 8237A-5.

RESET (Reset)

- Reset is an active high input which clears the command, status, request and temporary registers.
- It also clears the first/last flip/flop and sets the mask register.
- Following a reset the device is in the idle cycle.

A0-A3 (Address)

- The four least significant address lines are bi-directional three-state signals.
- In the idle cycle, they are inputs and are used by the 8237A to address the control register to be loaded or read.

- In the active cycle, they are outputs and provide the lower 4 bits of the output address.

CS (Chip select)

- The chip select is an active low input used to select the 8237A as an I/O device during the idle cycle.
- This allows CPU communication on the data bus.

Control logic and mode set registers

A4-A7 (Address)

- The four most significant address lines are three state outputs and provide 4 bits of the address.
- These lines enabled only during the DMA service.

READY (Ready)

- Ready is an input used to extend the memory read and write pulses from the 8237A to accommodate slow memories or I/O peripheral devices.

MEMW (Memory write)

The memory write is an active low three-state output used to write data to the selected memory location during a DMA read or memory-to-memory transfer.

AEN (Address enable)

- Address enable enables the 8-bit latch containing the upper 8 address bits onto the system address bus.
- AEN can also be used to disable other system bus drivers during DMA transfers.
- AEN is active HIGH.

ADSTB (Address strobe)

- The active high, address strobe is used to strobe the upper address byte into an external latch.

DREQ 0 – DREQ 3 (DMA request)

- The DMA request lines are individual asynchronous channel request inputs used by peripheral circuits to obtain DMA service.
- In fixed priority, DREQ 0 has the highest priority and DREQ 3 has the lowest priority.
- A request is generated by activating the DREQ line of a channel.
- DACK will acknowledge the recognition of DREQ signal.
- Polarity of DREQ is programmable.

- Reset initializes these lines to activate high.
- DREQ must be maintained until the corresponding DACK goes active.

DACK 0 – DACK 3 (DMA acknowledge)

- DMA acknowledge is used to notify the individual peripherals when one has been granted a DMA cycle.
- The sense of these lines is programmable.
- Reset initializes them to active low.

Registers in 8237

The 8237 has number of internal registers. These registers are used to control the DMA cycle operation.

1. Current address register

- Each channel contains a 16 bit current address register.
- This register holds the address of the data to be transferred.
- This address can be automatically incremented or decremented after each transfer.

2. Current word register

- Each channel has a 16 bit word count register.
- This register is used to determine the number of transfers to be performed.
- It is incremented after each byte transfer.

3. Base address register

- This 16 bit register stores the starting address of the current address register.

4. Base word register

- This 16 bit register stores the starting address of the current word register.

5. Command register

- This 8 bit register is used to program and control the 8237.
- It is used to initialize the device.

6. Mode register

- Each channel has a 8 bit mode register to define its mode of operation.

7. Request register

- This 4 bit register is used to request a DMA transfer by software.

8. Mask register

- Each channel has mask register bit that is used to disable incoming DREQ signals.

9. Temporary register

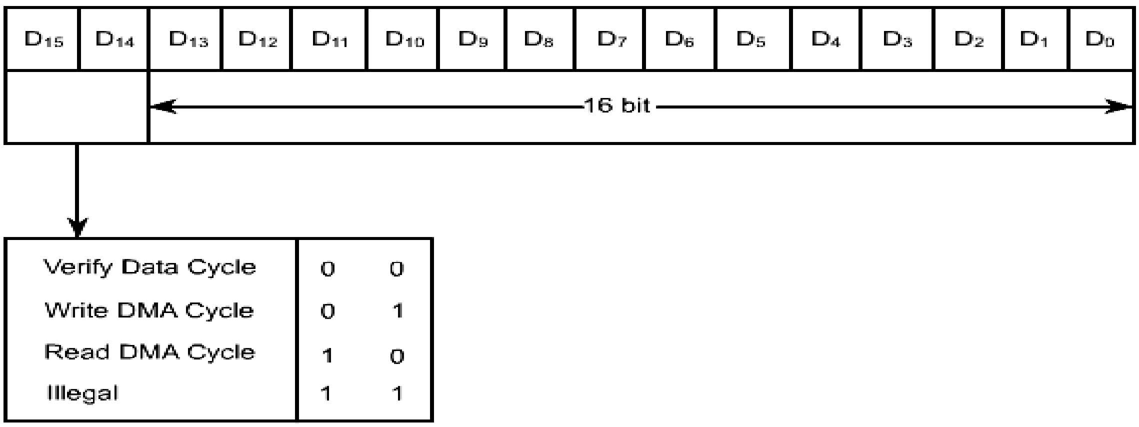
- This is an 8 bit register.
- It is used to hold the data during a memory-to-memory transfer.

10. Status register

- This is an 8 bit register.
- It is used for the microprocessor to read the present status of the 8237.

DMA channels

- The 8237 has four identical channels, each with two signals:
  - DRQ (DMA request)
  - DACK (DMA acknowledge)
- Each channel has two 16 bit register, one for the memory address where data transfer should begin, and the second for a 14 bit count.
- Bits D15 and D14 of the count register specify the DMA function such as write, read, or verify as shown in figure.
- In 8237 two additional 8 nit register are available to enable the DMA channels and to read the status of DMA channels.
- They are the control register, called the mode set register, shown in figure, and the status register shown in figure.
- The control word in the mode set register enables or disables channels and determines other functions.
- The port addresses of each register are determined by four address lines, as shown in figure and chip select logic.



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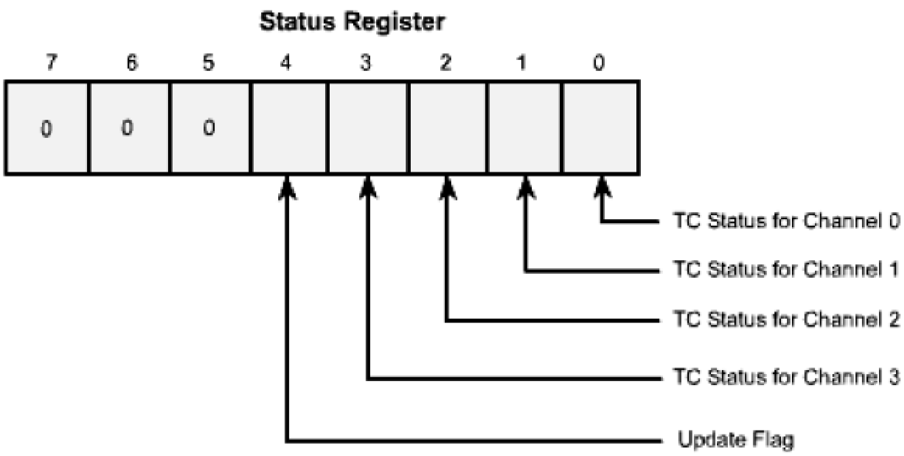
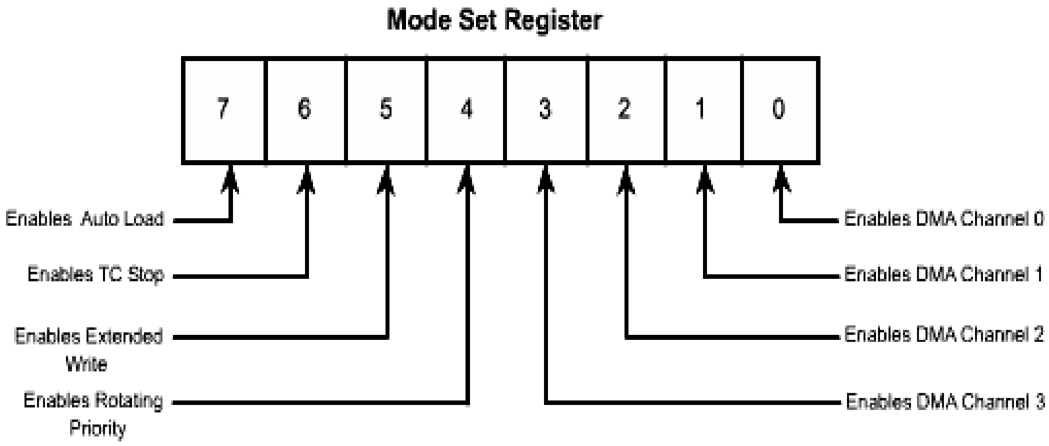
- The control word in the mode set register enables or disables channels and determines other functions.
- The port addresses of each register are determined by four address lines, as shown in figure and chip select logic.

A 3	A 2	A 1	A 0	
0	0	1	0	Selects channel 0 DMA address
0	0	0	1	Selects channel 0 terminate count address
1	0	0	0	Selects mode/status





Operation of DMA controller

- The 8237 DMA controller operates in two major cycles. They are

  i. The idle cycle.

  ii. The active cycle.

- When the 8237 is in the idle cycle, no external devices are requesting a DMA transfer.
- In this cycle, the 8237 samples the DREQ lines every clock cycle to determine if any of the four channels are requesting the service.
- When any one of the DMA request lines DREQ 0 – DREQ 3 becomes active, the 8237 will enter into the active cycle.
- First, the DMA controller requests a HOLD from the microprocessor in response to a DREQ (DMA request) from an external device.
- Next, when the microprocessor recognizes the request and is ready, it outputs a grant signal on the same signal line back to the DMA controller.
- The DMA controller can now use the buses to transfer data between memory and an external device
- The DMA controller now requests external device with a DACK signal.
- This indicates the start of a DMA transfer.
- After completing the data transfer by the DMA, it outputs an EOP, which indicates the completion of the DMA transfer.
- Then the control buses are returned to the microprocessor.

The 8237 in its active mode can perform a DMA transfer in any one of the following four modes:

Single transfer mode

- In this mode, 8237 transfers one byte of data each time the request is active.

Block transfer mode

- In this mode, 8237 transfer a block of data.
- The 8237 is programmed with the starting address of the data and the number of bytes to be transferred.
- The transfer is continued until the word count register in the 8237 reaches its final count or an EOP is received.

Demand transfer mode

- In this mode, the 8237 is programmed to continue making transfers until an EOP is received or until a DREQ from an external device goes inactive.
- In this mode, one byte of data is transferred for each demand (DREQ) received.
- Cascade transfer mode
- This mode is used to cascade more than one 8237 together of a system expansion.
- To use the DMA properly the programmer must first access the internal register of the 8237.
- The command register must be loaded with a command word.
- This command word defines the initial conditions, mode of operation, and the type of operation.

- Next, depending on the mode and the type of operation, the internal registers must be loaded with information defining the starting address of the data in memory, the number of bytes to be transferred, DMA channel number and so on.
- Finally, the DMA channel's request signal must be enabled.
- Placing control words does initialization and programming of the 8237.
- The OUT instruction is used to send control words to the internal registers.

DMA execution

The process of data transfer from the external devices to the system memory by the DMA controller can be classified under two modes:

- The slave mode
- The master mode

Slave mode

In the slave mode, the DMA controller is treated as a peripheral, using the following steps:

- Chip select is used to select the DMA.
- The microprocessor writes the command mode and terminal count in channel register by accessing the register through A0 – A3 and through the control signals IOR and IOW.

In this mode, the signals shown in the read / write logic block are used; the address line A7 – A4 and the control signals MEMR, MEMW from the control logic block are in tri-state. The other signals are not being used.

Master mode

After the initialization, the 8237 in master mode checks for a DMA request. The steps in data transfer can be listed as follows:

- When the peripheral is ready for data transfer, it sends a high signal to DRQ.
- When the DRQ has been received and the channel enabled, the control logic sets HRQ high.
- In the next cycle, the microprocessor releases the buses and sends the HLDA signal to the 8237.
- After receiving the HLDA signal, the control logic generates DACK (DMA acknowledge) and sends the acknowledgement to the peripheral.
- Meanwhile, the 8237 enables the signal AEN (address enable)
- AEN disables the microprocessor de-multiplexed address bus A7 – A0.
- The entire bus, A7 – A0, OF THE 8237 becomes output.
- The low-order byte of the memory location is placed on the A7-A0 of the 8237.
- When the AEN signal is high, the ADSTB (address strobe) signal goes high.

- This signal places the high-order byte of the memory location, generated by the lower 8212, on address bus A15-A8.
- Data transfer continues until the count reaches zero.

7. INTEL 8253 PROGRAMMABLE INTERVAL TIMER

- The 8253/54 solves one of the most common problem in any microcomputer system- the generation of accurate time delays under software control.
- Instead of setting up timing loops in system software, the programmer configures the 8253/54 with the desired quantity, then upon command the 8253/54 will count out the delay and interrupt the CPU when it has completed its tasks.

INTRODUCTION:

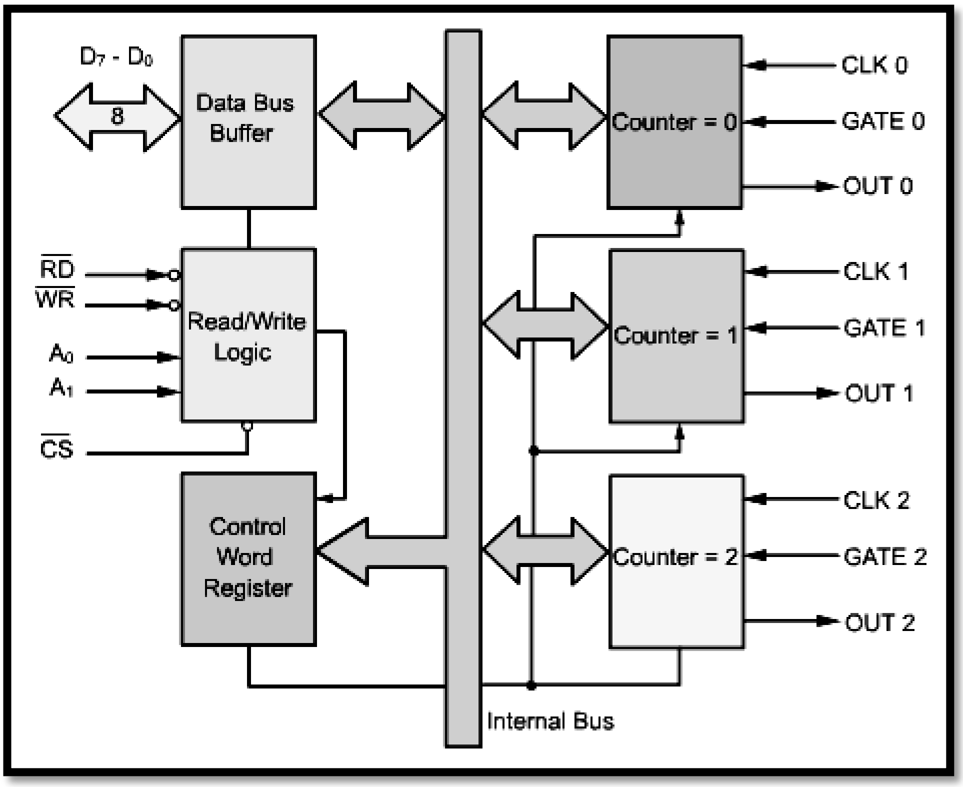
- The 8253 is a programmable interval timer/counter designed for use with intel microcomputer systems.
- It is a general purpose, multi-timing element that can be treated as an array of i/o ports in the system software.
- The 8253 solves one of the most common problems in any microcomputer system, the generation of accurate time delays under software control.
- Instead of setting up timing loops in software, the programmer configures the 8253 to match his requirements and programs one of the counters for the desired delay.
- After the desired delay, the 8253 will interrupt the cpu. Software overhead is minimal and variable length delays can easily be accommodated.
- The 8253/54 includes three identical 16 bit counters that can operate independently.
- To operate a counter, a 16 bit count is loaded in its register and, on command, it begins to decrement the count until it reaches 0.
- At the end of the count, it generates a pulse that can be used to interrupt the CPU.
- The counter can count either in binary or BCD.
- In addition, a count can be read by the CPU while the counter is decrementing.
- In this section, we are going to study two timer ICs 8253 and 8254 .
- The 8254 is a superset of 8253.

FEATURES:

- Three independent 16-bit down counters.
- 8254 can handle inputs from DC to 10 MHz (5 MHz 8254-5 8 MHz 8254 10 MHz 8254-2) where as 8253 can operate up to 2.6 MHz.
- Three counters are identical, pre-settable and can be programmed for either binary or BCD count.
- Counter can be programmed in six different modes.
- Compatible with all Intel and most other microprocessors.

- 8254 has powerful command called READ BACK command which allows the user to check the count value, programmed mode and current mode and current status of the counter.

Block diagram:



DATA BUS BUFFER:

- This tri-state, bi-directional, 8-bit buffer is used to interface the 8253/54 to the system data bus.
- The Data bus buffer has three basic functions.

    1. Programming the 8253/54 in various modes
    2. Loading the count registers.
    3. Reading the count values.

READ/WRITE LOGIC:

- The Read/Write logic has five signals: $\overline{RD}$, $\overline{WR}$, $\overline{CS}$ and the address lines A0 and A1.
- In peripheral I/O mode, the RD and WR signals are connected to IOR and IOW, respectively
- In memory-mapped I/O, these are connected to MEMR and MEMW. Address lines A0 and A1 of the CPU are usually connected to lines A0 and A1 of the CPU are

usually connected to lines A0 and A1 of the 8253/54, and CS is tied to a decoded address.

- The control word register and counters are selected according to the signals on lines A0 and A1.

Control word register:

- This register is accessed when lines A0 and A1 are at logic 1.
- It is used to a write a command word which specifies the counter to be used (binary or BCD), its mode and either a read or write operation.

A1	A0	SELECTION
0	0	COUNTER 0
0	1	COUNTER 1
1	0	COUNTER 2
1	1	CONTROL WORD REGISTER

Counter:

- These three functional blocks are identical in operation.
- Each counter consists of a single, 16 bit, pre-settable, down counter.
- The counter can operate in either binary or BCD and its input, gate and output are configured by the selection of modes stored in the control word register.
- The counters are fully independent.
- The programmer can read the contents of the three counters without disturbing the actual count in process.

Operational description:

- The complete functional definition of the 8254 is programmed by the system software.
- Once programmed the 8254 is ready to perform whatever timing tasks it is assigned to accomplish.

D7	D6	D5	D4	D3	D2	D1	D0
SC1	SC0	RW1	RW0	M2	M1	M0	BCD

- The register is accessed when lines A0 and A1 are at logic 1.
- It is used to write a command word which specifies the counter to tb used.
- Its mode & either a read or write operation.

SC1	SC0	
0	0	Select Counter 0
0	1	Select Counter 1
1	0	Select Counter 2
1	1	Read back command

The bits for D7 and D6 of the control word select anyone of the three counters or the read back command.

(SC-select counter D7 to D6)

The bits D5 and D4 decide the various read/write operation.

RW1	RW2	
0	0	Counter latch command
0	1	Read/write least significant byte only
1	0	Read/write most significant byte only
1	1	Read/write least significant byte first, then most significant byte

The bits D3, D2 & D1 decide the mode in which the 8254 has to operate the six modes.

M2	M1	M0	
0	0	0	Mode 0
0	0	1	Mode 1
X	1	0	Mode 2
X	1	1	Mode 3
1	0	0	Mode 4
1	0	1	Mode 5

M-Mode(D3,D2 &D1)

The bits D0 of the control word decides whether counter should count in BCD or Binary.

BCD	
0	Binary counter 16-bits
1	Binary coded decimal(BCD)counter(4 decades)

Modes of operation

The 8254 can operate in six different modes, and the gate of a counter is used either to disable or enable counting as given below.

Signal status / Modes	Low or Going Low	Rising	High
0	Disables counting	----	Enables counting
1	----	1.Initiates counting 2. Resets output after next clock	----
2	1. Disables counting 2. Sets output immediately high.	1. Reloads counter 2. Initiates counting	Enables counting
3	1. Disables counting 2. Sets output immediately high.	Initiates counting	Enables counting
4	Disables counting	----	Enables counting
5	----	Initiates counting	----

<u>Mode Definition:</u>

READ AND WRITE OPERATIONS

The 8254 can be programmed to provide various types of output through write operations, or to check a count while counting through read operations. The details of these operations are given below.

READ OPERATIONS

In some applications especially in event counters, it is necessary to read the values of the count in progress. This can be done by either of two methods.

- One method involves reading a count after inhibiting (stopping) the counter to be read. It is known as reading by halting the count.
- The second method involves reading a count while the count is in progress (known as reading on the fly). It is known as reading while counting.

In first method, counting is stopped (or inhibited) by controlling the gate input or the clock input of the selected counter, and two I/O read operations are performed by the MPU. The first I/O operation reads the low order byte, and the second I/O operation reads the higher order byte.

In the second method, an appropriate control word is written into the control register to latch a count in the output latch, and two I/O read operations are performed by the MPU.

WRITE OPERATION:

To initialize a counter, the following steps are necessary.

- Write a control word into the control register.
- Load the low order byte of a count in the counter register.
- Load the high order byte of a count in the counter register.

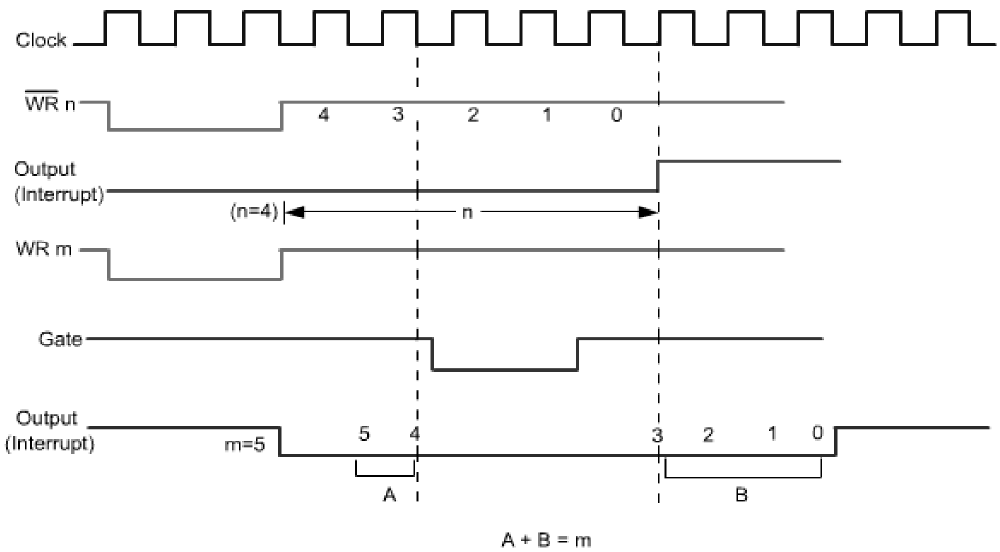
With a clock and an appropriate gate signal to one of the counters, the above steps should start the counter and provide appropriate output according to the control word.

MODES OF OPERATION:

As mentioned earlier, the 8254 can operate in six different modes; we will describe briefly various modes of the 8254.

MODE 0: Interrupt on terminal count

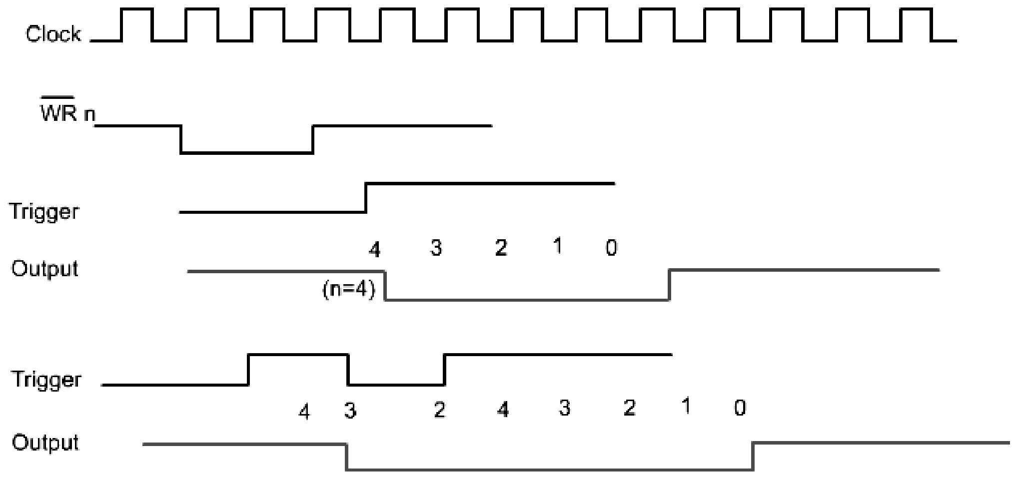
- The output goes low on setting the mode, and goes high after the desired count.
- In this mode, initially the OUT is low.
- Once count is loaded in the register, the counter is decremented every cycle, and when the count reaches zero, the OUT goes high.
- This can be used as an interrupt.



- The OUT remains high until a new count or a command word is loaded.
- Figure shows that the counting (m=s) is temporarily stopped when the gate is disabled (G=0), and continued again when the gate is at login 1.

MODE 1: Hardware-Retriggerable One-Shot

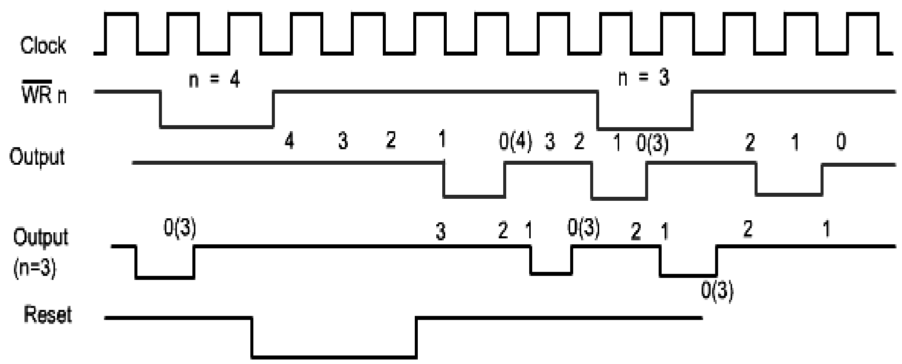
- The output goes low on a gate input, and goes high on terminal count.



- In this mode, the OUT is initially high.
- When the gate is triggered, the out goes low, and at the end of the count, the OUT goes high again, thus generating a one-shot pulse.

MODE 2: Rate generator

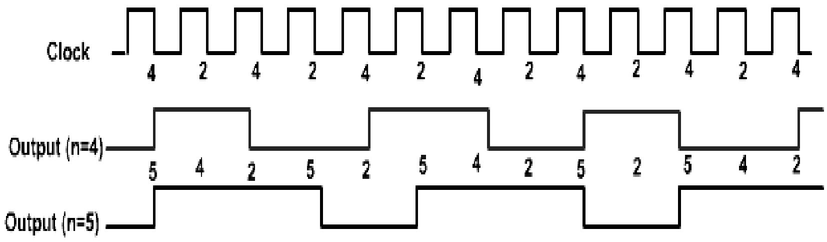
- It is equivalent to a ‘clock pulse divide by n’ counter.
- The output pulse frequency is 1/n of the input pulse frequency ,where n is the count number .
- This mode is used to generate pulse equal to the clock period at a given interval
- When a count is loaded, the OUT stays high until the count reaches 1, and then the OUT goes low for one clock Period.



- The count is reloaded automatically, and the pulse is generated continuously.
- The count = 1 is illegal in this mode.

MODE 3: Square wave generator

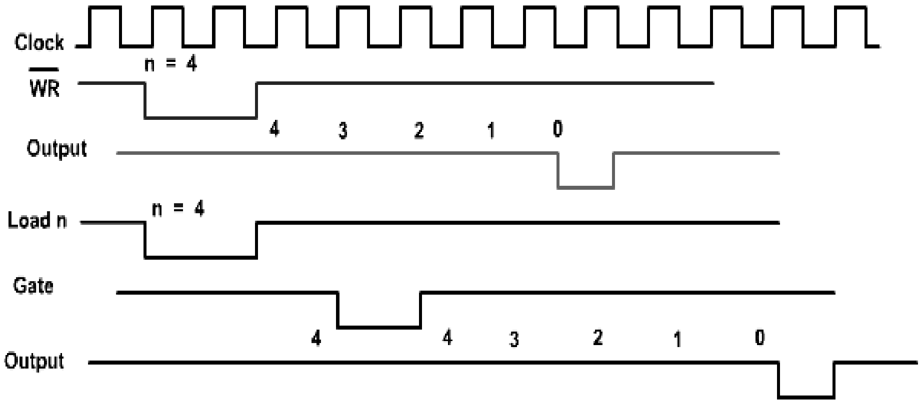
- This is similar to mode 2 with difference in minor operating details.
- In this mode, when a count is loaded, the OUT is high.
- The count is decremented by two at every clock cycle, and when it reaches zero, the OUT goes low, and the count is reloaded again.



- This is repeated continuously; thus, a continuous square wave with period equal to the period of the count is generated.
- In other word, the frequency of the square wave is equal to the frequency of the clock divided by the count.
- If the count (N) is odd the pulse stays high for (N+1)/2 clock cycles and stays low for (N-1)/2 clock cycles.

MODE 4: Software-triggered strobe

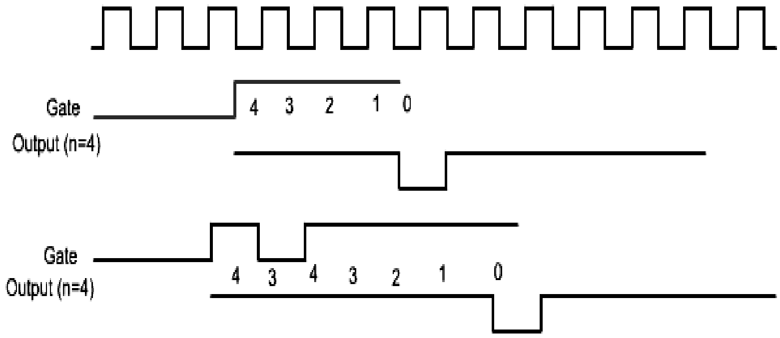
- The output goes high on setting the mode
- After terminal count, the output goes low for one clock period and then goes high again



- In this mode, the OUT is initially high; it goes low for one clock period at the end of the count.
- The count must be reloaded for subsequent outputs.

MODE 5: Hardware-triggered strobe

- This is similar to mode 4, but a trigger at the gate initiates the counting.
- This mode is similar to mode 4, except that it is triggered by the rising pulse at the gate.

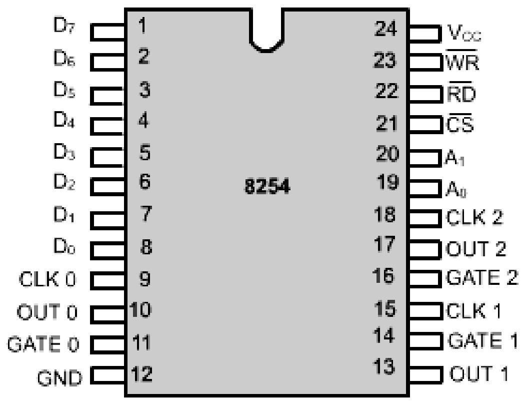


- Initially, the OUT is high, and when the gate pulse is triggered from low to high, the count begins.
- At the end of the count, the OUT goes low for one clock period.

Difference between 8253A and 8254A:

Sl.no	8253	8254
1	Operating frequency 0-26 MHz.	Operating frequency 0-10 MHz.
2	Uses N-MOS technology	Uses H-MOS technology
3	Read-back command not available	Read-back command available
4	Reads and writes of the same counter cannot be interleaved	Reads and writes of the same counter can be interleaved

PIN DIAGRAM:



D_7 - D_0	Data Bus (Bidirectional)
CLK N	Counter Clock Inputs
GATE N	Counter Gate Inputs
OUT N	Counter Outputs
$\overline{RD}$	Read Counter
$\overline{WR}$	Write Command or Data
$\overline{CS}$	Chip Select
A_0 - A_1	Counter Select
V_{CC}	+ 5 volts
GND	0 Volts

Pin diagram details:

Symbol	Pin No.	Type	Name and Function
D_7-D_0	1–8	I/O	**DATA:** Bi-directional three state data bus lines, connected to system data bus.
CLK 0	9	I	**CLOCK 0:** Clock input of Counter 0.
OUT 0	10	O	**OUTPUT 0:** Output of Counter 0.
GATE 0	11	I	**GATE 0:** Gate input of Counter 0.
GND	12		**GROUND:** Power supply connection.
V_{CC}	24		**POWER:** +5V power supply connection.
$\overline{WR}$	23	I	**WRITE CONTROL:** This input is low during CPU write operations.
$\overline{RD}$	22	I	**READ CONTROL:** This input is low during CPU read operations.
$\overline{CS}$	21	I	**CHIP SELECT:** A low on this input enables the 8254 to respond to $\overline{RD}$ and $\overline{WR}$ signals. $\overline{RD}$ and $\overline{WR}$ are ignored otherwise.
A_1, A_0	20–19	I	**ADDRESS:** Used to select one of the three Counters or the Control Word Register for read or write operations. Normally connected to the system address bus. **A_1** — **A_0** — **Selects** 0 — 0 — Counter 0 0 — 1 — Counter 1 1 — 0 — Counter 2 1 — 1 — Control Word Register
CLK 2	18	I	**CLOCK 2:** Clock input of Counter 2.
OUT 2	17	O	**OUT 2:** Output of Counter 2.
GATE 2	16	I	**GATE 2:** Gate input of Counter 2.
CLK 1	15	I	**CLOCK 1:** Clock input of Counter 1.
GATE 1	14	I	**GATE 1:** Gate input of Counter 1.
OUT 1	13	O	**OUT 1:** Output of Counter 1.